

SHOFIA FAUZIYAH

Course by NordEdu
SQL #3

15 DAYS OF USING
SQL

Mastering Data Retrieval & Management with
PostgreSQL

Challenge

SYNTAX

```
1  -- In the email system there was a problem with names where
2  -- either the first name or the last name is more than 10 characters
3  -- long.
4  -- Find these customers and output the list of these first and last
5  -- names in all lower case.
6  SELECT
7      LOWER(first_name) AS first_name,
8      LOWER(last_name) AS last_name,
9      email
10 FROM customer
11 WHERE LENGTH(first_name) > 10
12        OR LENGTH(last_name) > 10;
```

OUTPUT

Data Output Messages Notifications			
	first_name text	last_name text	email text
1	william	satterfield	WILLIAM.SATTERFIELD@sakilacustomer.org
2	christopher	greco	CHRISTOPHER.GRECO@sakilacustomer.org
3	henry	billingsley	HENRY.BILLINGSLEY@sakilacustomer.org
4	roger	quintanilla	ROGER.QUINTANILLA@sakilacustomer.org
5	jonathan	scarborough	JONATHAN.SCARBOROUGH@sakilacustomer.o...
6	allen	butterfield	ALLEN.BUTTERFIELD@sakilacustomer.org
7	mittchell	westmorela...	MITCHELL.WESTMORELAND@sakilacustomer....
8	nelson	christenson	NELSON.CHRISTENSON@sakilacustomer.org
Total rows: 8		Query complete 00:00:00.402	

notes

Terdapat 8 customer dari total 599 yang memenuhi kondisi nama depan atau nama belakang lebih dari 10 karakter dalam format huruf kecil beserta emailnya

Untuk memastikan kembali bisa dengan menambahkan kolom length agar terlihat jumlah/total huruf pada setiap nama yang memenuhi kondisi

Data Output Messages Notifications					
	first_name text	first_name_length integer	last_name text	last_name_length integer	email text
1	william	7	satterfield	11	WILLIAM.SATTERFIELD@sakilacustomer.org
2	christopher	11	greco	5	CHRISTOPHER.GRECO@sakilacustomer.org
3	henry	5	billingsley	11	HENRY.BILLINGSLEY@sakilacustomer.org
4	roger	5	quintanilla	11	ROGER.QUINTANILLA@sakilacustomer.org
5	jonathan	8	scarborough	11	JONATHAN.SCARBOROUGH@sakilacustomer.o...
6	allen	5	butterfield	11	ALLEN.BUTTERFIELD@sakilacustomer.org
7	mittchell	8	westmorela...	12	MITCHELL.WESTMORELAND@sakilacustomer....
8	nelson	6	christenson	11	NELSON.CHRISTENSON@sakilacustomer.org

Challenge

SYNTAX

```
24 -- In this challenge you have only the email address and the last
25 -- name of the customers.
26 -- You need to extract the first name from the email address and
27 -- concatenate it with the last name. It should be in the form:
28 -- "Last name, First name".
29 SELECT
30     last_name || ', ' ||
31     LEFT(email, POSITION('.') IN email) - 1) AS customer_name
32 FROM customer;
```

notes

Menampilkan nama customer dalam format "Last name dan First name" yang diekstrak dari email dan kolom last_name

OUTPUT

Data Output

Messages

Notifications

customer_name

text

1

SMITH, MARY

2

JOHNSON, PATRICIA

3

WILLIAMS, LINDA

4

JONES, BARBARA

5

BROWN, ELIZABETH

6

DAVIS, JENNIFER

7

MILLER, MARIA

8

WILSON, SUSAN

Challenge

SYNTAX

```
34 -- Extract the last 5 characters of the email address first.  
35 -- The email address always ends with '.org'.  
36 -- How can you extract just the dot '.' from the email address?  
37 SELECT  
38     RIGHT(email, 5) AS right,  
39     SUBSTRING(RIGHT(email, 5), 2, 1) AS left  
40 FROM customer;
```

notes

Menampilkan 5 karakter terakhir email (r.org) dan karakter titik (.) yang diekstrak dari email customer

OUTPUT

Data Output		Message
	right text 	left text 
1	r.org	.
2	r.org	.
3	r.org	.
4	r.org	.
5	r.org	.

Challenge

SYNTAX

```
50 -- You need to create an anonymized form of the email addresses
51 -- in the following way:
52 SELECT
53     LEFT(email, 1) || '***.' || SUBSTRING(email, POSITION('.') IN email) + 1, 1)
54     || '***' || SUBSTRING(email, POSITION('@' IN email))
55     AS anonymized_email,
56     email
57 FROM customer;
```

OUTPUT

Data Output Messages Notifications		
Showing rows		
	anonymized_email text	email text
1	M***.S***@sakilacustomer....	MARY.SMITH@sakilacustomer.org
2	P***.J***@sakilacustomer.o...	PATRICIA.JOHNSON@sakilacustomer.org
3	L***.W***@sakilacustomer....	LINDA.WILLIAMS@sakilacustomer.org
4	B***.J***@sakilacustomer.o...	BARBARA.JONES@sakilacustomer.org
5	E***.B***@sakilacustomer.o...	ELIZABETH.BROWN@sakilacustomer.org
6	J***.D***@sakilacustomer.o...	JENNIFER.DAVIS@sakilacustomer.org
7	M***.M***@sakilacustomer....	MARIA.MILLER@sakilacustomer.org

notes

Menampilkan email yang telah di anonymized atau disamarkan dengan format Huruf Pertama pada Nama*. Huruf Pertama pada LastName***@domain"*, dan domain email tetap terlihat

Challenge

SYNTAX

```
59 -- In a second query create an anonymized form of the email
60 -- addresses in the following way:
61 SELECT
62     '***' || SUBSTRING(email, POSITION('.') IN email) - 1, 1)
63     || '.' || SUBSTRING(email, POSITION('.') IN email) + 1, 1)
64     || '***' || SUBSTRING(email, POSITION('@' IN email))
65     AS anonymized_email,
66     email
67 FROM customer;
```

OUTPUT

Data OutputMessagesNotifications

Showing rows:

	anonymized_email text	email text
1	***Y.S***@sakilacustomer.o...	MARY.SMITH@sakilacustomer.org
2	***A.J***@sakilacustomer.o...	PATRICIA.JOHNSON@sakilacustomer.org
3	***A.W***@sakilacustomer....	LINDA.WILLIAMS@sakilacustomer.org
4	***A.J***@sakilacustomer.o...	BARBARA.JONES@sakilacustomer.org
5	***H.B***@sakilacustomer....	ELIZABETH.BROWN@sakilacustomer.org

notes

Menampilkan email yang telah di anonymized atau disamarkan dan domain email tetap terlihat

Challenge

SYNTAX

```
69 -- You need to analyze the payments and find out the following:
70 -- What's the month with the highest total payment amount?
71 SELECT
72     EXTRACT(MONTH FROM payment_date) AS month,
73     SUM(amount) AS total_payment_amount
74 FROM payment
75 GROUP BY EXTRACT(MONTH FROM payment_date)
76 ORDER BY total_payment_amount DESC;
```

OUTPUT

Data Output			Messages	Notifications
	month numeric	total_payment_amount numeric		
1	4	27226.52		
2	3	23886.56		
3	2	9745.61		
4	1	4710.70		
5	5	1847.12		

notes

Dari output didapatkan total pembayaran tertinggi sebesar 28.378,02 pada bulan april

Challenge

SYNTAX

```
78 -- What's the day of week with the highest total payment amount? (0 is Sunday)
79 SELECT
80     EXTRACT(DOW FROM payment_date) AS day_of_week,
81     SUM(amount) AS total_payment_amount
82 FROM payment
83 GROUP BY EXTRACT(DOW FROM payment_date)
84 ORDER BY total_payment_amount DESC;
```

OUTPUT

	Data Output	Messages	Notifications
	day_of_week numeric		total_payment_amount numeric
1	4		12354.45
2	1		12042.29
3	2		9897.63
4	3		9772.99
5	0		8878.70

notes

Dari output didapatkan hasil hari Kamis (Thursday) memiliki total pembayaran tertinggi sebesar 12.796,08,

Challenge

```
97 -- You need to sum payments and group in the following formats.
98 SELECT
99     SUM(amount) AS total_amount,
100     TO_CHAR(payment_date, 'Dy, DD/MM/YYYY') AS day
101 FROM payment
102 GROUP BY TO_CHAR(payment_date, 'Dy, DD/MM/YYYY')
103 ORDER BY total_amount;
```

Data Output Messages Notifications

Showing rows: 1 to 41

	total_amount numeric	day text
1	113.73	Sat, 01/02/2020
2	181.58	Sat, 22/02/2020
3	328.27	Mon, 13/04/2020

notes

Format dengan mengambil inisial atau singkatan hari, lalu membuat format date menggunakan /

Challenge

```
105 -- You need to sum payments and group in the following formats:
106 SELECT
107     SUM(amount) AS total_amount,
108     TO_CHAR(payment_date, 'Mon, YYYY') AS day
109 FROM payment
110 GROUP BY TO_CHAR(payment_date, 'Mon, YYYY')
111 ORDER BY total_amount;
```

Data Output Messages Notifications

	total_amount numeric	day text
1	1847.12	May, 2020
2	4710.70	Jan, 2020
3	9745.61	Feb, 2020
4	23886.56	Mar, 2020
5	27226.52	Apr, 2020

notes

Format dengan mengambil inisial atau singkatan hari, menampilkan hanya day dan tahun

Challenge

```
113 -- You need to sum payments and group in the following formats.
114 SELECT
115     SUM(amount) AS total_amount,
116     TO_CHAR(payment_date, 'Dy, HH24:MI') AS day
117 FROM payment
118 GROUP BY TO_CHAR(payment_date, 'Dy, HH24:MI')
119 ORDER BY total_amount DESC;
120
```

Data Output Messages Notifications

	total_amount numeric	day text
1	537.14	Thu, 19:44
2	45.92	Wed, 09:17
3	43.93	Thu, 23:54

notes

Format dengan mengambil inisial atau singkatan hari, menampilkan hanya day dan time

Challenge For Today
#DAY3

Challenge #1

SYNTAX

```
135 -- Also you need to find out for the suppcity team
136 -- which customer has the longest average rental duration?
137 SELECT
138     customer_id,
139     AVG(return_date - rental_date) AS avg_interval
140 FROM rental
141 GROUP BY customer_id
142 ORDER BY avg_interval DESC;
```

OUTPUT

Data Output			Messages	Notifications
	customer_id smallint	avg_interval interval		
1	315	6 days 14:13:22.5		
2	187	5 days 34:58:38.5714...		
3	321	5 days 32:56:32.7272...		
4	539	5 days 31:39:57.2727...		
5	436	5 days 31:09:46		
6	532	5 days 30:59:34.8387...		
7	427	5 days 29:27:05		
8	555	5 days 26:48:35.2941...		
Total rows: 599			Query complete 00:00:00	

notes

Customer_id 315 memiliki highest rental duration dengan durasi nya selama 6 days 14:13:22.5

Challenge #2

SYNTAX

```
134 -- You need to create a list for the suppcity team of all rental
135 -- durations of customer with customer_id 35.
136 SELECT
137     customer_id,
138     return_date - rental_date AS rental_duration
139 FROM rental
140 WHERE customer_id = 35;
```

OUTPUT

Data Output

Messages

Notifications

	customer_id smallint	rental_duration interval
1	35	4 days 20:59:00
2	35	8 days 18:10:00
3	35	5 days 01:12:00
4	35	1 day 18:25:00
5	35	8 days 03:26:00
6	35	5 days 03:00:00
7	35	3 days 20:15:00
8	35	5 days 22:10:00

Total rows: 32

Query complete 00:00:00

notes

Terdapat 32 history rental date dari customer_id 35 dengan highest rental duration selama 4 days 20:59:00

THANKYOU